

Title: Database Normalization Report

Project: Football Tournament Management System

1. Introduction

This document demonstrates that the football tournament database is designed according to the principles of the **3rd Normal Form (3NF)** to ensure data integrity and minimize redundancy.

2. Normalization Process

First Normal Form (1NF)

No repeating groups, and all values are atomic. In our database, every column contains only a single value, and there are no multi-valued attributes.

- **Demonstration:** Every table in our database (e.g., Players, Teams, Matches) contains only one value per cell. For example, the Players table doesn't list multiple positions for one player in a single cell; each position is a single string. There are no repeating columns like goal1, goal2, etc.

Second Normal Form (2NF)

Criteria: Must be in 1NF AND have no partial dependencies (every non-key attribute must depend on the *entire* primary key).

- **Demonstration:** In tables with a single-column Primary Key (like Stadiums), 2NF is automatically satisfied.
- **Composite Keys:** In our bridge tables like MatchPlayers(match_id, player_id), the attribute goals_scored depends on **both** the match and the player. You cannot know the goals scored without knowing both IDs.

Third Normal Form (3NF)

Criteria: Must be in 2NF AND have no transitive dependencies (non-key attributes should not depend on other non-key attributes).

- **Demonstration:** * In the Teams table, country depends directly on team_id.
 - In the Stadiums table, location and capacity depend only on the stadium_id.
 - We avoided putting the Stadium Location directly into the Matches table; instead, we use a stadium_id to reference it. This prevents a transitive dependency where Location would depend on Stadium, which depends on Match.

3. Table-by-Table Summary

Table	PK	Functional Dependencies (FDs)	Normal Form
Players	player_id	player_id $\rightarrow$ name, position, team_id	3NF
Matches	match_id	match_id $\rightarrow$ match_date, stadium_id, tournament_id	3NF
MatchPlayers	(match_id, player_id)	(match_id, player_id) $\rightarrow$ goals_scored	3NF
Stadiums	stadium_id	stadium_id $\rightarrow$ name, location, capacity	3NF

4. Denormalization

No denormalization was performed for this project. The database maintains full normalization to ensure maximum data consistency and to prevent anomalies during INSERT or UPDATE operations. The current scale of the database does not require denormalization for performance optimization.

5. Conclusion

By strictly following the 1NF, 2NF, and 3NF rules, the database is robust, eliminates redundant data, and ensures that all relationships between entities (Players, Teams, Sponsors, etc.) are logically sound.